

Chapter 5: Elasticity and Its Application

Chapter Objectives (1 of 3)

By the end of this chapter, you should be able to:

- Calculate price elasticity of demand in a given scenario.
- Identify regions of the demand curve as elastic, inelastic, or unit elastic using price elasticity of demand.
- List the factors that influence price elasticity of demand.
- Describe the relationship between price elasticity of demand and the slope of a demand curve.
- Explain how changes in supply impact market equilibrium.

Chapter Objectives (2 of 3)

- Analyze the relationship between price elasticity of demand and total revenue.
- Calculate income elasticity of demand in a given scenario.
- Determine whether a good is inferior or normal using income elasticity of demand.
- Calculate cross-price elasticity of demand in a given scenario.
- Determine if two goods are complements or substitutes using cross-price elasticity of demand.
- Calculate price elasticity of supply in a given scenario.
- List the factors that influence price elasticity of supply.

Chapter Objectives (3 of 3)

- Identify regions of the supply curve as elastic, inelastic, or unit elastic using price elasticity of supply.
- Explain how changes in demand impact market equilibrium.

5.1 The Elasticity of Demand

The Price Elasticity of Demand

- **Elasticity***

- Measure of the responsiveness of quantity demanded or quantity supplied to a change in one of its determinants

- **Price elasticity of demand***

- Measures how much the quantity demanded responds to a change in the price

*Words accompanied by an asterisk are key terms from the chapter.

Elastic and Inelastic Demand

- Elastic demand
 - Quantity demanded responds substantially to price changes
- Inelastic demand
 - Quantity demanded responds only slightly to price changes

Determinants of Price Elasticity of Demand (1 of 2)

- Availability of close substitutes
 - Elastic demand: Goods with close substitutes
 - Inelastic demand: Goods with no close substitute
- Necessities and luxuries
 - Elastic demand: Luxuries
 - Inelastic demand: Necessities

Determinants of Price Elasticity of Demand

(2 of 2)

- Defining the market broadly or narrowly
 - Elastic demand: Narrowly defined markets
 - Inelastic demand: Broadly defined markets
- Time horizon
 - More elastic demand: Longer time horizons
 - Less elastic demand: Shorter time horizons

Computing the Price Elasticity of Demand

- Computing the price elasticity of demand
 - Percentage change in quantity demanded divided by percentage change in price
 - Use absolute value (drop the minus sign)

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

The Midpoint Method

- The standard procedure for calculating a percentage change is to divide the change by the initial level
- The midpoint method divides the change by the midpoint (or average) of the initial and final levels

$$\text{Price elasticity of demand} = \frac{(Q_2 - Q_1)[(Q_2 + Q_1) / 2]}{(P_2 - P_1)[(P_2 + P_1) / 2]}$$

Active Learning 1: Calculate an Elasticity

Use the following information to calculate the price elasticity of demand for iPhones:

- $P = \$400$, $Q_d = 10,600$
- $P = \$600$, $Q_d = 8,400$

- A. Use the midpoint method to calculate percentage change in price
- B. Use the midpoint method to calculate percentage change in quantity
- C. Calculate the price elasticity of demand

Active Learning 1: Answers

Using the midpoint method to calculate percentage changes:

- A. % change in $P = [(\$600 - \$400)/\$500] \times 100 = 40\%$
- B. % change in $Q_d = [(8,400 - 10,600)/9,500] \times 100 = -23.16\%$
- C. Price elasticity of demand
= % change in Q_d / % change in P
= $23.16/40 = 0.58$ (ignoring the minus sign)

The Variety of Demand Curves (1 of 2)

- Demand is elastic
 - Price elasticity of demand > 1
- Demand is inelastic
 - Price elasticity of demand < 1
- Demand has unit elasticity
 - Price elasticity of demand $= 1$

The Variety of Demand Curves (2 of 2)

- Demand is perfectly inelastic
 - Price elasticity of demand = 0
 - Demand curve is vertical
- Demand is perfectly elastic
 - Price elasticity of demand = infinity
 - Demand curve is horizontal

Figure 1 The Price Elasticity of Demand (1 of 2)

The price elasticity of demand determines whether the demand curve is steep or flat. Note that all percentage changes are calculated using the midpoint method.

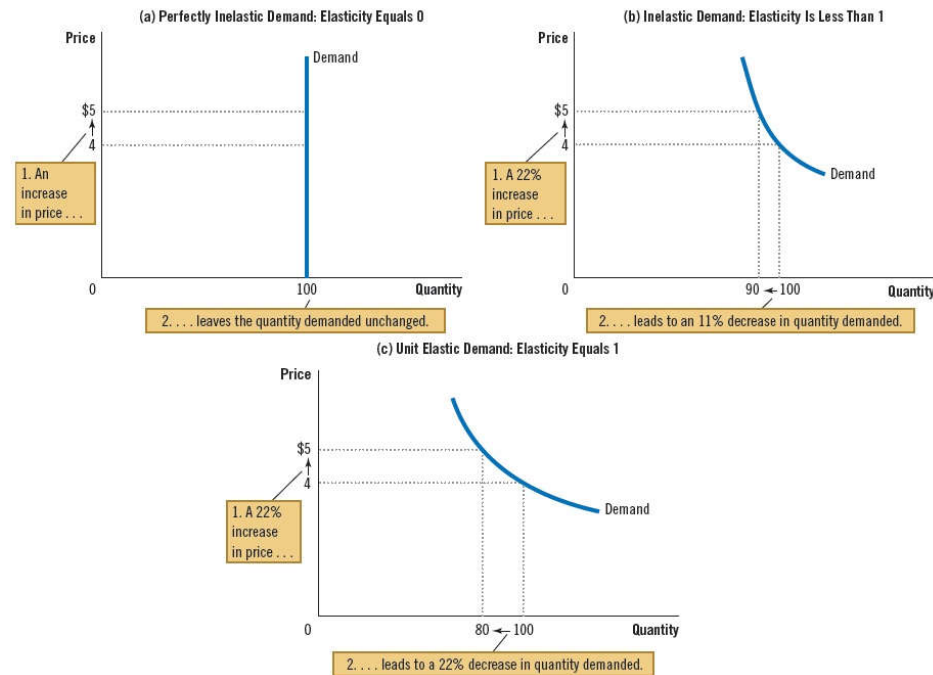
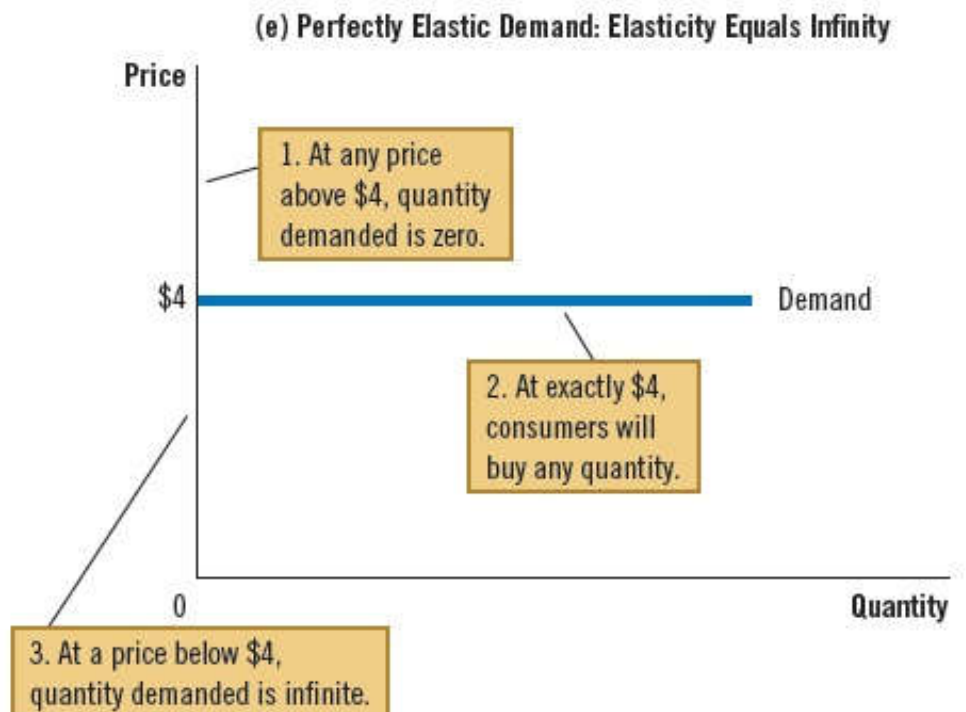
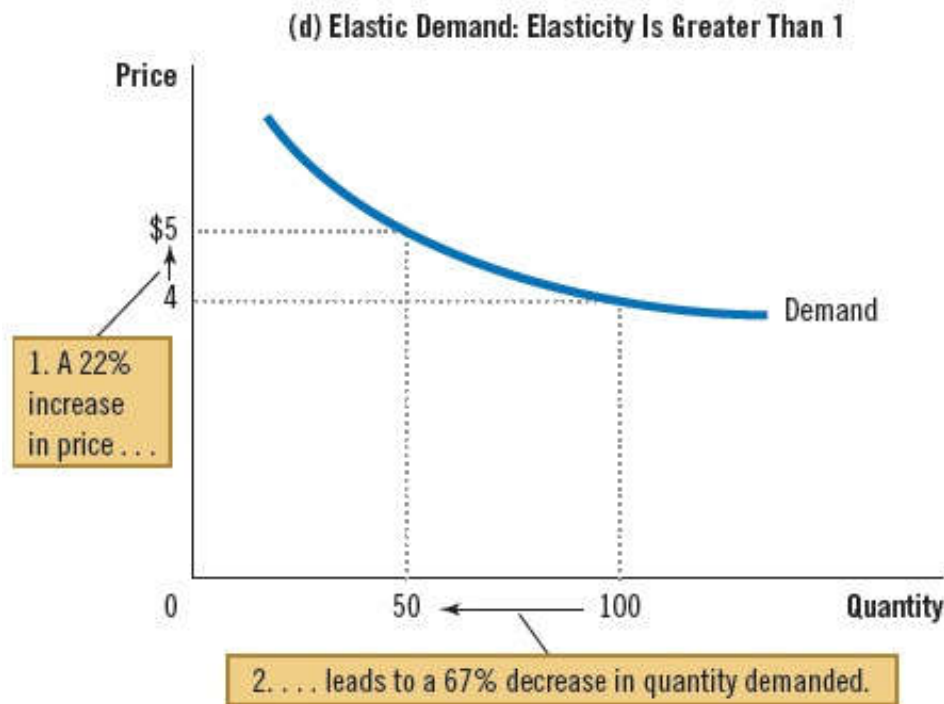


Figure 1 The Price Elasticity of Demand (2 of 2)



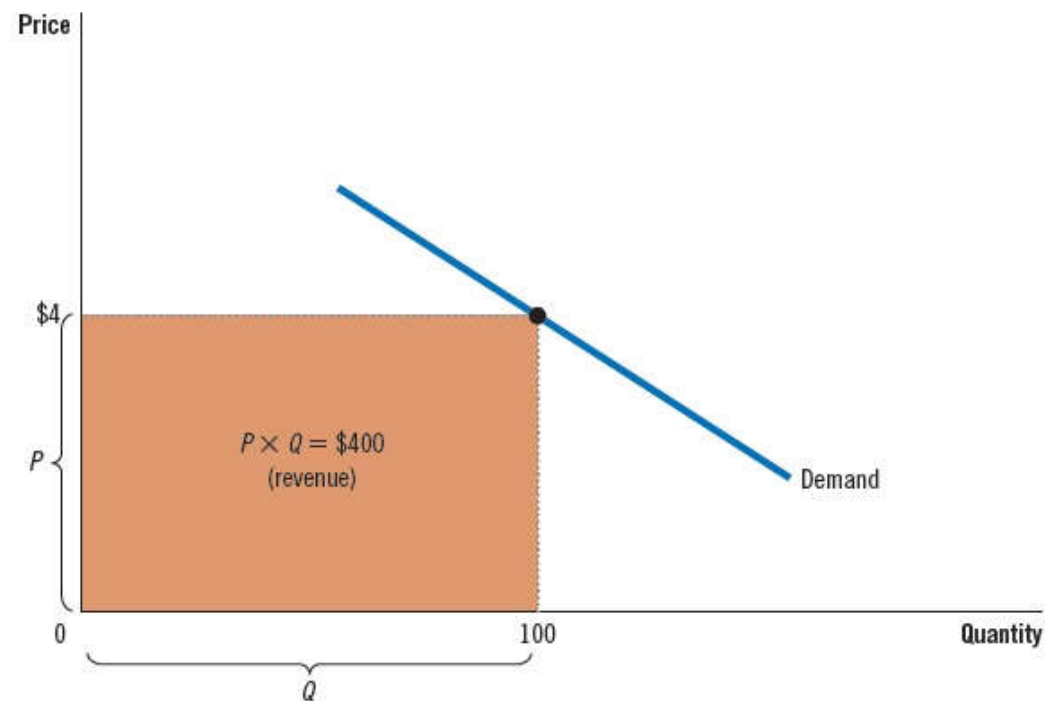
Total Revenue

- **Total revenue***
 - Amount paid by buyers and received by sellers of a good
 - Price of the good times the quantity sold ($P \times Q$)

*Words accompanied by an asterisk are key terms from the chapter.

Figure 2 Total Revenue

- The area of the box under the demand curve, $P \times Q$, equals the total amount paid by buyers as well as the total revenue received by sellers.
- Here, at a price of \$4, the quantity demanded is 100, and total revenue is \$400.

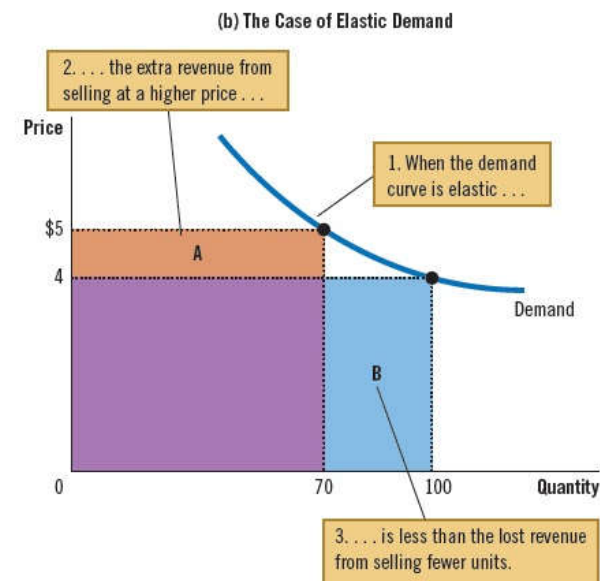
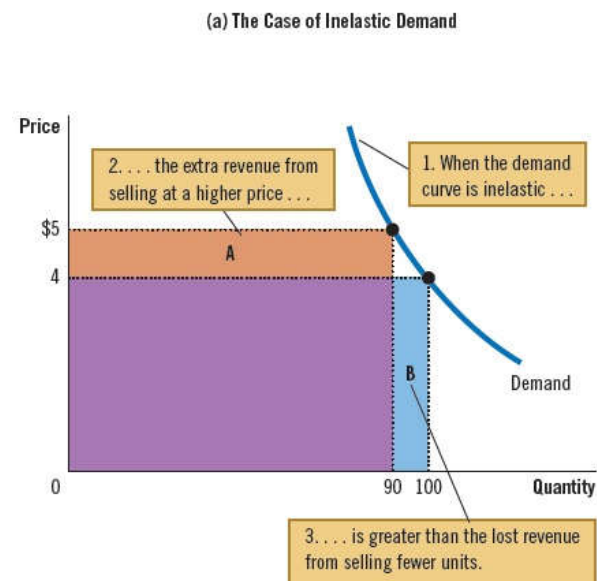


Total Revenue and the Price Elasticity of Demand

- Elastic demand
 - Price and total revenue move in opposite directions
- Inelastic demand
 - Price and total revenue move in the same direction
- Unit elastic demand
 - When price changes, total revenue remains constant

Figure 3 How Total Revenue Changes When Price Changes

The impact of a price change on total revenue (price times quantity) depends on the elasticity of demand. In panel (a), the demand curve is inelastic. A price increase leads to a proportionately smaller decrease in quantity demanded, so total revenue increases. Here, the price increases from \$4 to \$5, and the quantity demanded falls from 100 to 90. Total revenue rises from \$400 to \$450. In panel (b), the demand curve is elastic. A price increase leads to a proportionately larger decrease in quantity demanded, so total revenue decreases. Here, the price increases from \$4 to \$5, and the quantity demanded falls from 100 to 70. Total revenue falls from \$400 to \$350.



Elasticity and Total Revenue along a Linear Demand Curve

- If demand curve is linear, it has a constant slope
 - “Rise over run”
- Elasticity changes
 - Elastic demand: Points with high price and low quantity
 - Inelastic demand: Points with low price and high quantity

Figure 4 Elasticity along a Linear Demand Curve (1 of 2)

- The slope of a linear demand curve is constant, but its elasticity is not.
- The price elasticity of demand is calculated using the demand schedule and the midpoint method.
- At points with a low price and high quantity, the demand curve is inelastic.
- At points with a high price and low quantity, it is elastic.

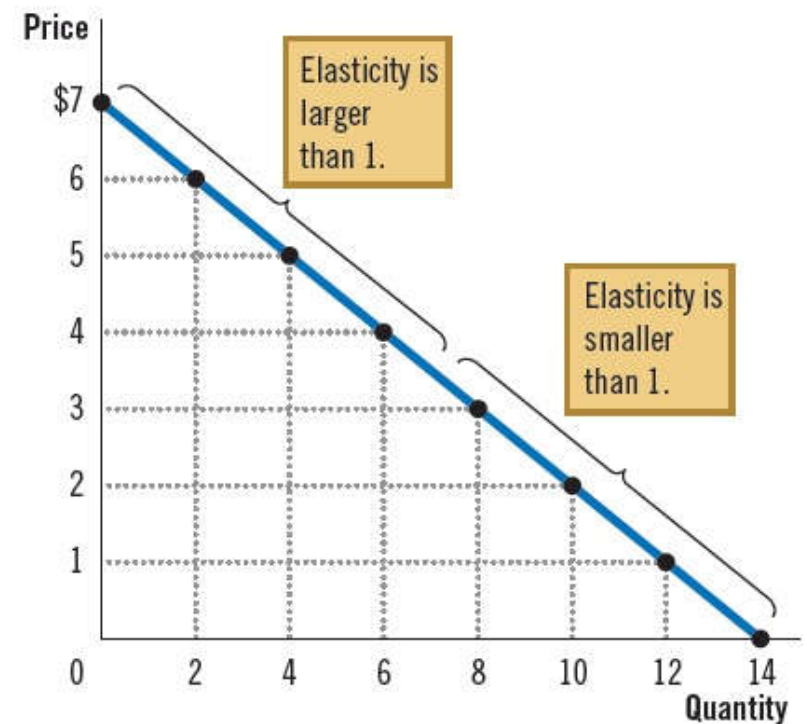


Figure 4 Elasticity along a Linear Demand Curve (2 of 2)

Price	Quantity	Total Revenue (Price × Quantity)	Percentage Change in Price	Percentage Change in Quantity	Elasticity	Description
\$7	0	\$0				
6	2	12	15	200	13.0	Elastic
5	4	20	18	67	3.7	Elastic
4	6	24	22	40	1.8	Elastic
3	8	24	29	29	1.0	Unit elastic
2	10	20	40	22	0.6	Inelastic
1	12	12	67	18	0.3	Inelastic
0	14	0	200	15	0.1	Inelastic

Active Learning 2: Elasticity and Total Revenue

- A. Pharmacies raise the price of insulin by 10%. Does total expenditure on insulin rise or fall?
- B. As a result of a fare war, the price of a luxury cruise falls 20%. Does luxury cruise companies' total revenue rise or fall?

Active Learning 2: Answers

A. Expenditure = Total Revenue = $P \times Q$

- Insulin is a necessity
- Since demand for insulin is inelastic, Q will fall less than 10%, so expenditure rises

B. Revenue = $P \times Q$

- The fall in P reduces revenue, but Q increases, which increases revenue
- Since demand is elastic, Q will increase more than 20%, so revenue rises

The Income Elasticity of Demand

- **Income elasticity of demand***

- A measure of how much the quantity demanded of a good responds to a change in consumers' income
 - Normal goods have a positive income elasticity
 - Inferior goods have a negative income elasticity

$$\text{Income elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

*Words accompanied by an asterisk are key terms from the chapter.

The Cross-Price Elasticity of Demand

- **Cross-price elasticity of demand***

- A measure of how much the quantity demanded of one good responds to a change in the price of another good
 - Substitutes have a positive cross-price elasticity
 - Complements have a negative cross-price elasticity

$$\text{Cross-price elasticity of demand} = \frac{\text{Percentage change in quantity demanded of good one}}{\text{Percentage change in the price of good two}}$$

*Words accompanied by an asterisk are key terms from the chapter.

5.2 The Elasticity of Supply

The Price Elasticity of Supply

- **Price elasticity of supply***
 - Measures how much the quantity supplied of a good responds to a change in the price of that good

*Words accompanied by an asterisk are key terms from the chapter.

Elastic and Inelastic Supply

- Elastic supply
 - Quantity supplied responds substantially to price changes
- Inelastic supply
 - Quantity supplied responds only slightly to price changes

Determinants of Price Elasticity of Supply

- Depends on the flexibility of sellers to change the amount they produce
- Short run
 - Quantity supplied is not very responsive to changes in price
- Long run
 - Quantity supplied responds substantially to price changes

Computing the Price Elasticity of Supply

- Computing the price elasticity of demand
 - Percentage change in quantity supplied divided by percentage change in price
 - Use midpoint method

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

The Variety of Supply Curves (1 of 2)

- Supply is elastic
 - Price elasticity of supply > 1
- Supply is inelastic
 - Price elasticity of supply < 1
- Supply has unit elasticity
 - Price elasticity of supply $= 1$

The Variety of Supply Curves (2 of 2)

- Supply is perfectly inelastic
 - Price elasticity of supply = 0
 - Supply curve is vertical
- Supply is perfectly elastic
 - Price elasticity of supply = infinity
 - Supply curve is horizontal

Figure 5 The Price Elasticity of Supply (1 of 3)

The price elasticity of supply determines whether the supply curve is steep or flat. Note that all percentage changes are calculated using the midpoint method.

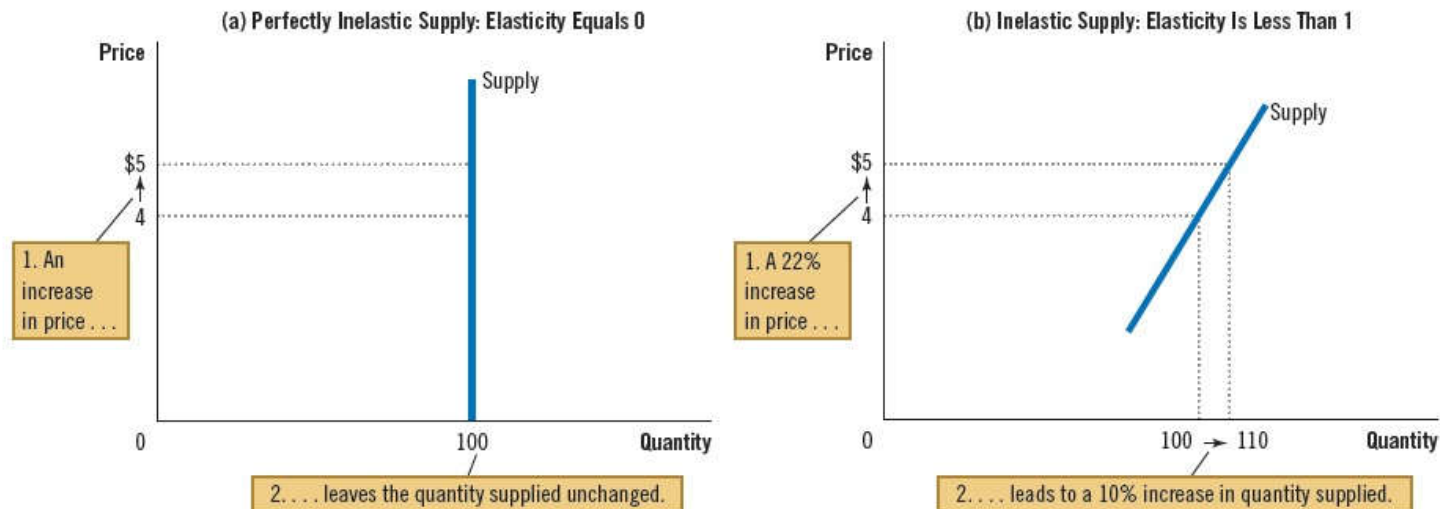


Figure 5 The Price Elasticity of Supply (2 of 3)

The price elasticity of supply determines whether the supply curve is steep or flat. Note that all percentage changes are calculated using the midpoint method.

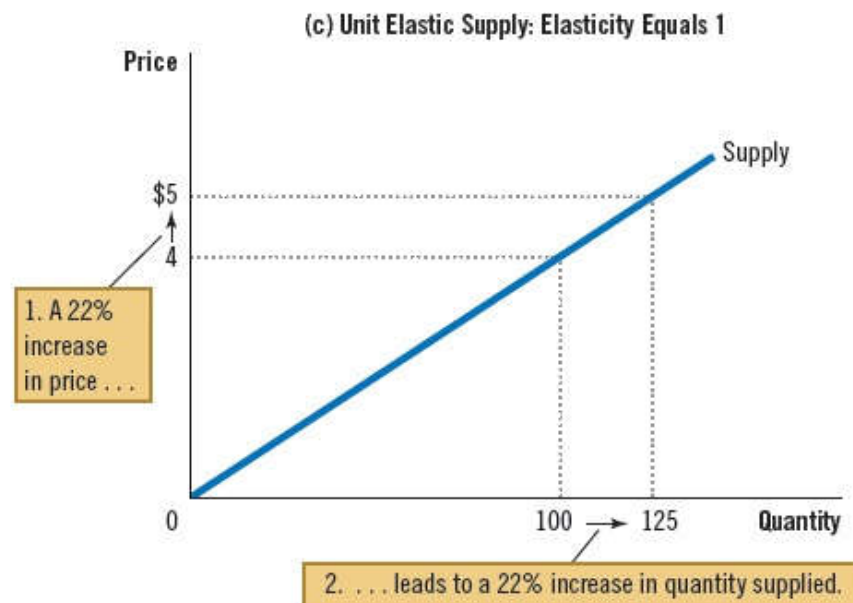


Figure 5 The Price Elasticity of Supply (3 of 3)

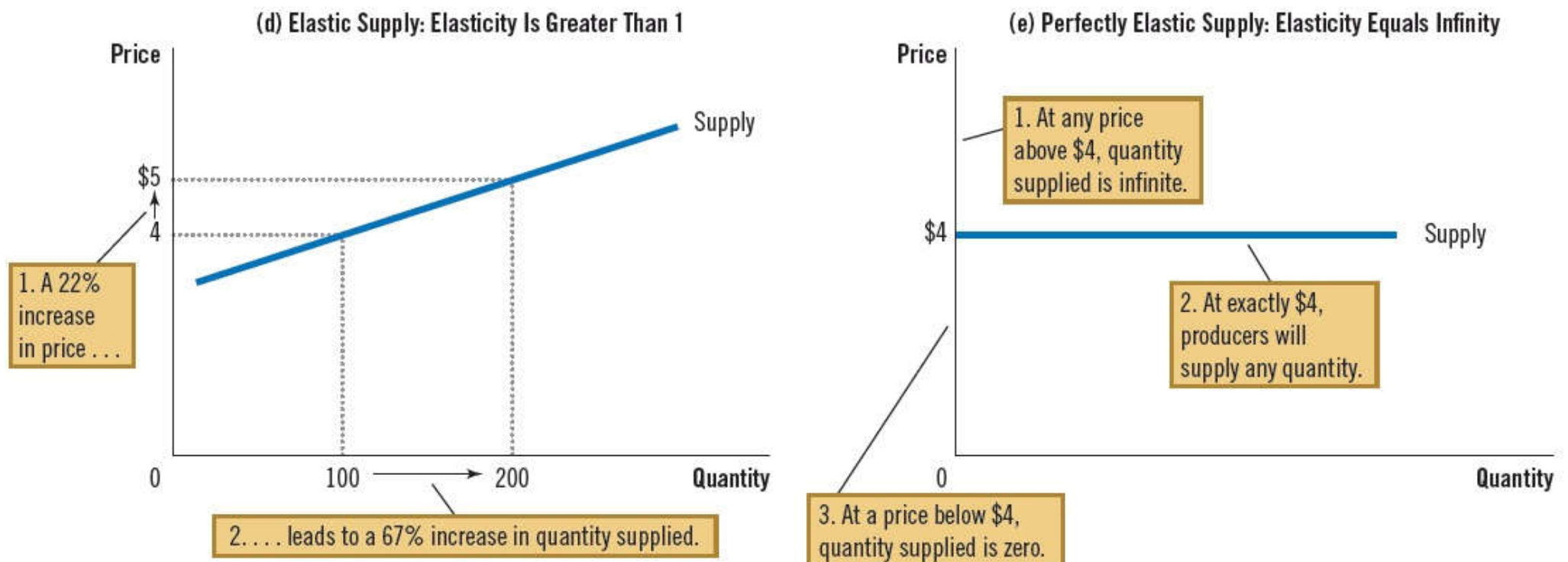
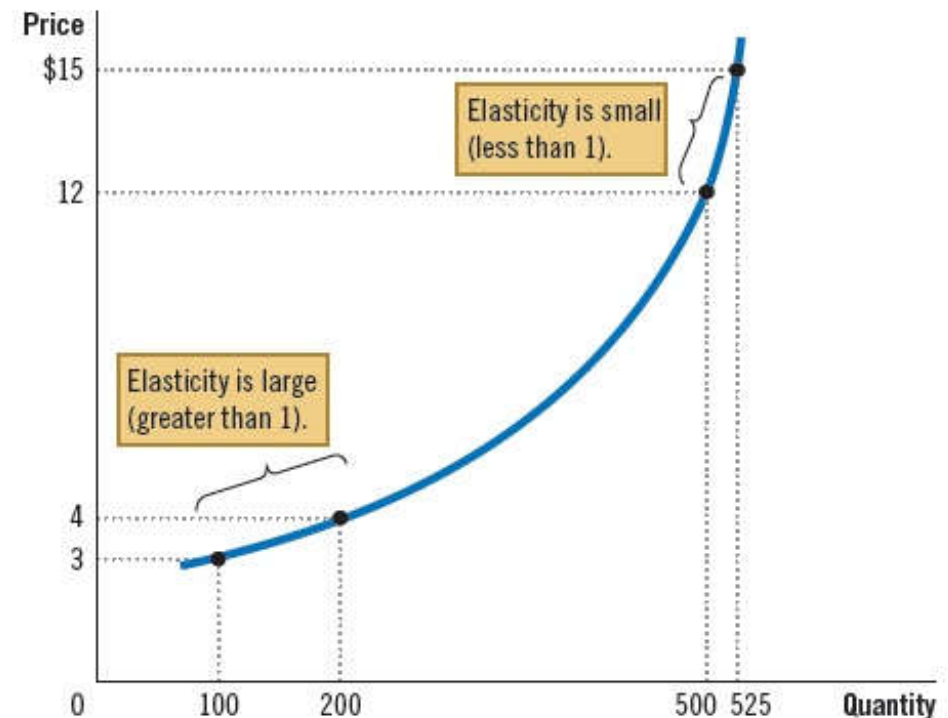


Figure 6 How the Price Elasticity of Supply Can Vary

- Because firms often have a maximum capacity for production, the elasticity of supply may be very high at low levels of quantity supplied and very low at high levels of quantity supplied.
- Here, an increase in the price from \$3 to \$4 increases the quantity supplied from 100 to 200. Because the 67% increase in quantity supplied (calculated with the midpoint method) is larger than the 29% increase in price, the supply curve in this range is elastic.
- By contrast, when the price rises from \$12 to \$15, the quantity supplied rises only from 500 to 525. Because the 5 percent increase in quantity supplied is smaller than the 22% increase in price, the supply curve in this range is inelastic.



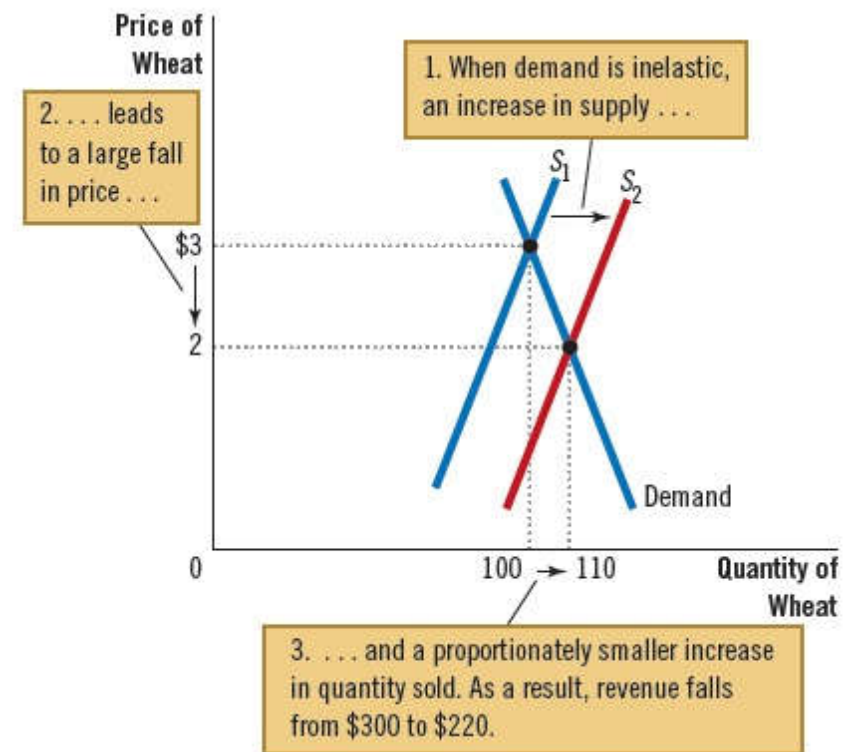
5.3 Three Applications of Supply, Demand, and Elasticity

Can Good News for Farming Be Bad News for Farmers?

- New hybrid of wheat increases production per acre by 20%
 - Supply curve shifts to the right
 - Higher quantity and lower price
 - Demand is inelastic: Total revenue falls
- Paradox of public policy
 - Induce farmers not to plant crops

Figure 7 A Supply Increase in the Market for Wheat

- When a technological advance increases the wheat supply from S_1 to S_2 , the price falls.
- Because the demand for wheat is inelastic, the increase in quantity from 100 to 110 is proportionately smaller than the decrease in the price from \$3 to \$2.
- As a result, farmers' total revenue falls from \$300 ($\3×100) to \$220 ($\2×110).

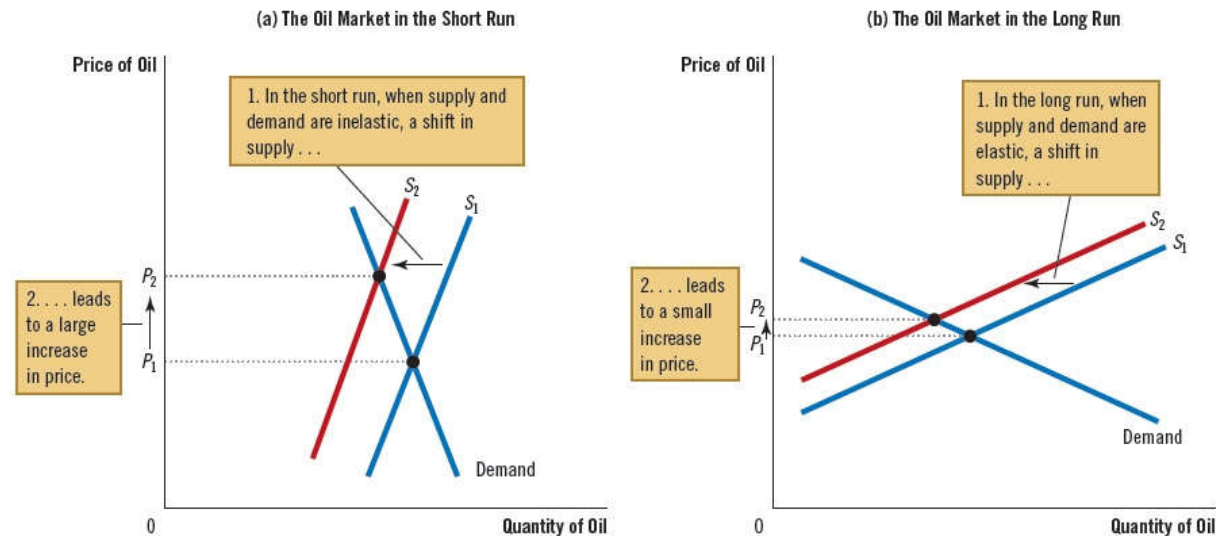


Why Has OPEC Failed to Keep the Price of Oil High?

- Increase in prices: 1973–1974, 1971–1981
- Short-run: Supply and demand are inelastic
 - Decrease in supply: Large increase in price
- Long-run: Supply and demand are elastic
 - Decrease in supply: Small increase in price

Figure 8 A Reduction in Supply in the World Market for Oil

When the supply of oil falls, the response depends on the time horizon. In the short run, supply and demand are relatively inelastic, as in panel (a). The shift in the supply curve from S_1 to S_2 leads to a substantial price increase. In the long run, however, supply and demand are relatively elastic, as in panel (b). In this case, the same size shift in the supply curve (S_1 to S_2) causes a smaller price increase.



Does Drug Interdiction Increase or Decrease Drug-Related Crime? (1 of 2)

- Increase the number of federal agents devoted to the war on drugs
 - Supply curve shifts left
 - Lower quantity and higher price
 - Demand is inelastic: Total revenue rises

Does Drug Interdiction Increase or Decrease Drug-Related Crime? (2 of 2)

- Drug interdiction would increase drug-related crime in the short run but decrease it in the long run
 - Demand is probably inelastic over short periods because higher prices do not substantially affect drug use by addicts
 - Demand may be more elastic over longer periods because higher prices would discourage experimentation and lead to fewer drug addicts

5.4 Conclusion

- The tools of supply and demand are useful for analyzing the events and policies that shape the economy

Think-Pair-Share Activity

In order to reduce teen smoking, the government places a \$2 per pack tax on cigarettes. After one month, the quantity demanded of cigarettes has been reduced only slightly. Discuss the following:

- A. What conclusion can you draw about the one-month demand for cigarettes?
- B. Caleb suggests that the cigarette industry should get together and raise the price of cigarettes further to increase total revenue
- C. Keisha suggests that only your firm should raise the price of your cigarettes to increase total revenue

Chapter 6: Supply, Demand, and Government Policies

Chapter Objectives (1 of 2)

By the end of this chapter, you should be able to:

- Determine the impact of price controls on economic welfare using the supply and demand model.
- Determine if a price control is a price ceiling or a price floor using the supply and demand model.
- Determine if a price control is binding using the supply and demand model.
- Determine the amount of shortage or surplus generated by a price control using the supply and demand model.

Chapter Objectives (2 of 2)

- Describe the unintended consequences of rent control using the supply and demand model.
- Explain how a change in a labor supply determinant impacts labor supply.
- Identify the tax incidence on consumers and producers for a given market.
- Determine the impact of a tax on the equilibrium price and quantity in a market.
- Analyze the relationship between elasticity and tax burden.

6.1 The Surprising Effects of Price Controls

Price Controls

- Economists as policy analysts and advisers try to use theories to change the world
- Policymakers often enact price controls when they believe that the market price of a good or service is too high or too low
- These policies can generate problems of their own

Price Ceiling and Price Floor

- Price ceiling*

- A legal maximum on the price at which a good can be sold
 - Rent-control laws

- Price floor*

- A legal minimum on the price at which a good can be sold
 - Minimum wage laws

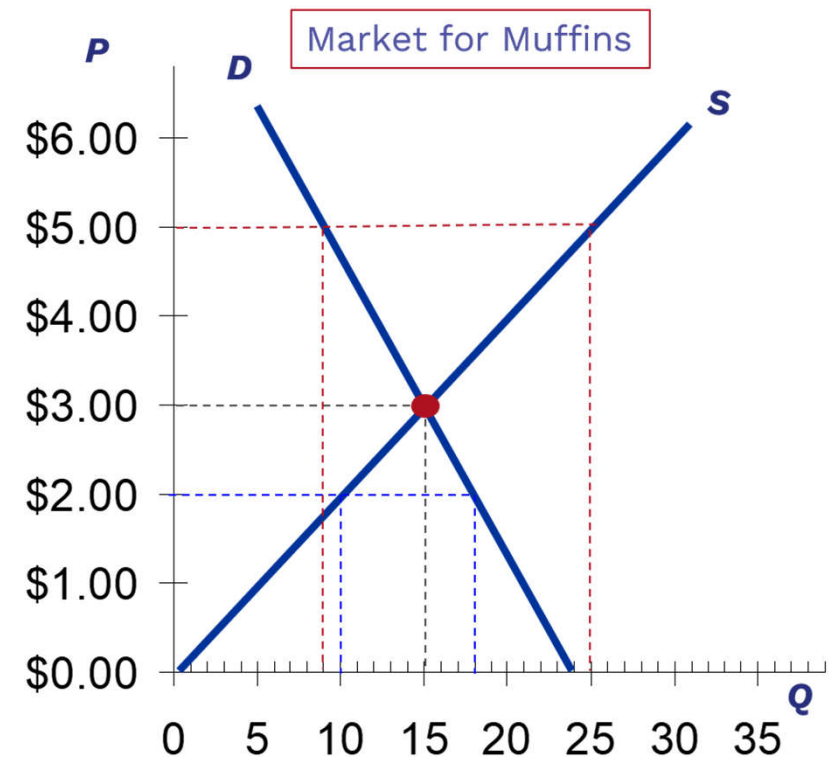
*Words accompanied by an asterisk are key terms from the chapter.

How Price Ceilings Affect Market Outcomes

- Not binding
 - Set above the equilibrium price
 - No effect on price or quantity sold
- Binding constraint
 - Set below the equilibrium price
 - Market price must be the price ceiling

Active Learning 1: Price Ceilings for Muffins

- The Muffin Buyers' Association lobbies the government to impose a price ceiling. Which of the following is binding and what's the effect on the market?
- The price ceiling is set at \$5
- The price ceiling is set at \$2



Active Learning 1: Answers

A. The price ceiling is set at \$5

- Not binding
- $P = \$3, Q = 15$

B. The price ceiling is set at \$2

- Binding
- $P = \$2, Q_d = 18, Q_s = 10$
- Shortage = 8 muffins

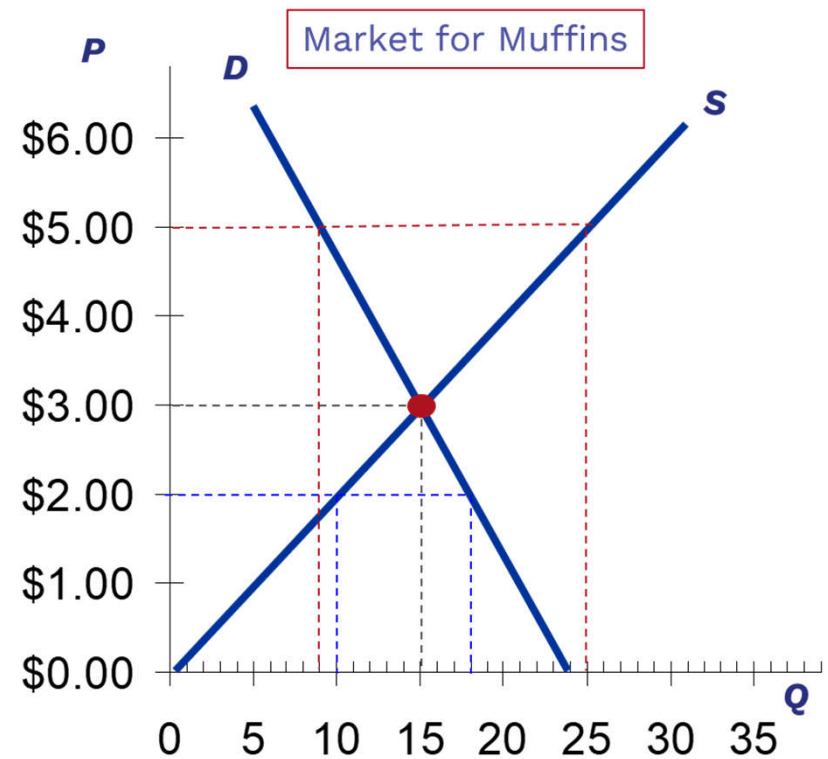
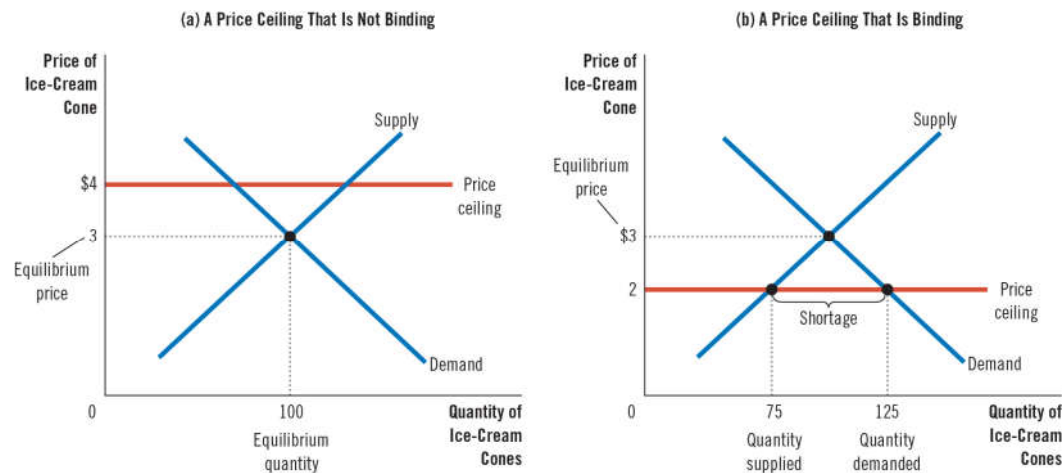


Figure 1 A Market with a Price Ceiling

In panel (a), the government imposes a price ceiling of \$4. Because it is above the equilibrium price of \$3, the ceiling has no effect, and the market can reach the equilibrium of supply and demand. At this point, quantity supplied and quantity demanded both equal 100 cones. In panel (b), the government imposes a price ceiling of \$2. Because the ceiling is below the equilibrium price of \$3, the market price is \$2. At this price, 125 cones are demanded while only 75 are supplied, so there is a shortage of 50 cones.



Binding Price Ceiling

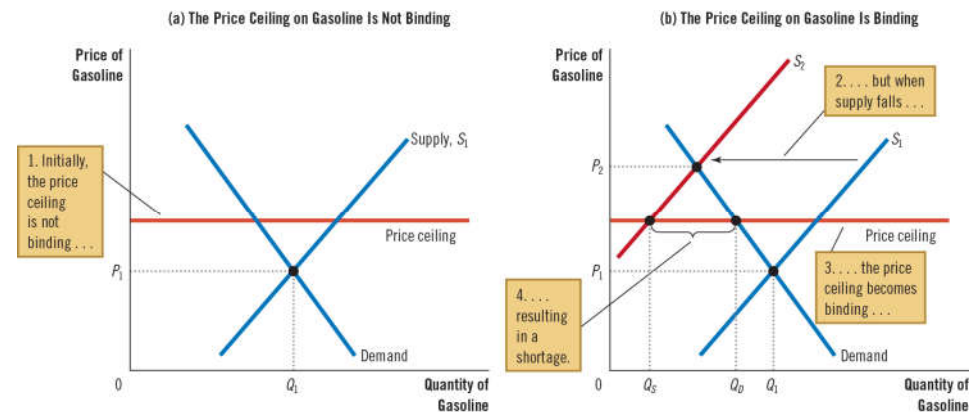
- When the government imposes a binding price ceiling
 - Shortage arises
 - Sellers must ration scarce goods among potential buyers
- Rationing mechanisms are rarely desirable
 - Long lines waste buyers' time
 - Bias of sellers
 - Inefficient (good may not go to the buyer who values it most)
 - Unfair

Rationing Mechanism

- Rationing mechanism in a free, competitive market is straightforward
 - When market reaches its equilibrium, anyone who wants to pay the market price can buy the good
 - May seem unfair to some buyers when prices are high
 - Is efficient and impersonal

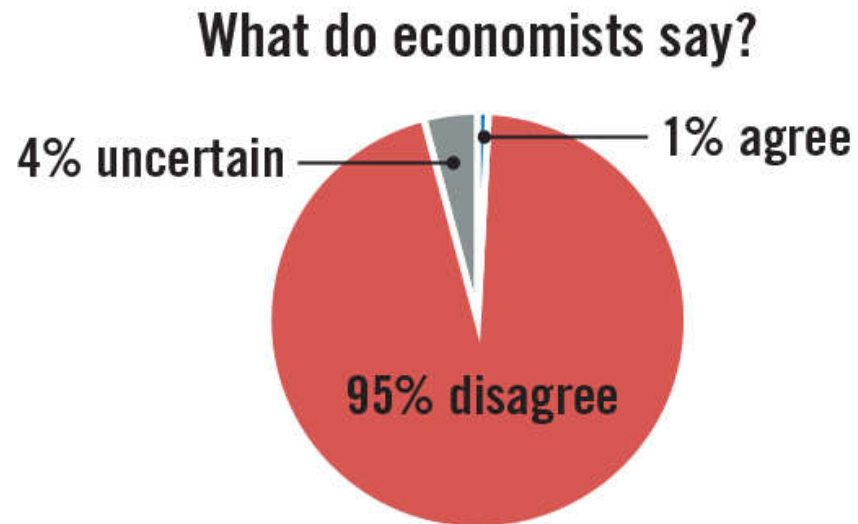
Figure 2 The Market for Gasoline with a Price Ceiling

Panel (a) shows the gasoline market when the price ceiling is not binding because the equilibrium price, P_1 , is below the ceiling. Panel (b) shows the gasoline market after an increase in the price of crude oil (an input into making gasoline) shifts the supply curve to the left from S_1 to S_2 . In an unregulated market, the price would have risen from P_1 to P_2 . The price ceiling, however, prevents this from happening. At the binding price ceiling, consumers are willing to buy Q_D , but producers of gasoline are willing to sell only Q_S . The difference between quantity demanded and quantity supplied, $Q_D - Q_S$, measures the gasoline shortage.



Ask the Experts: Rent Control

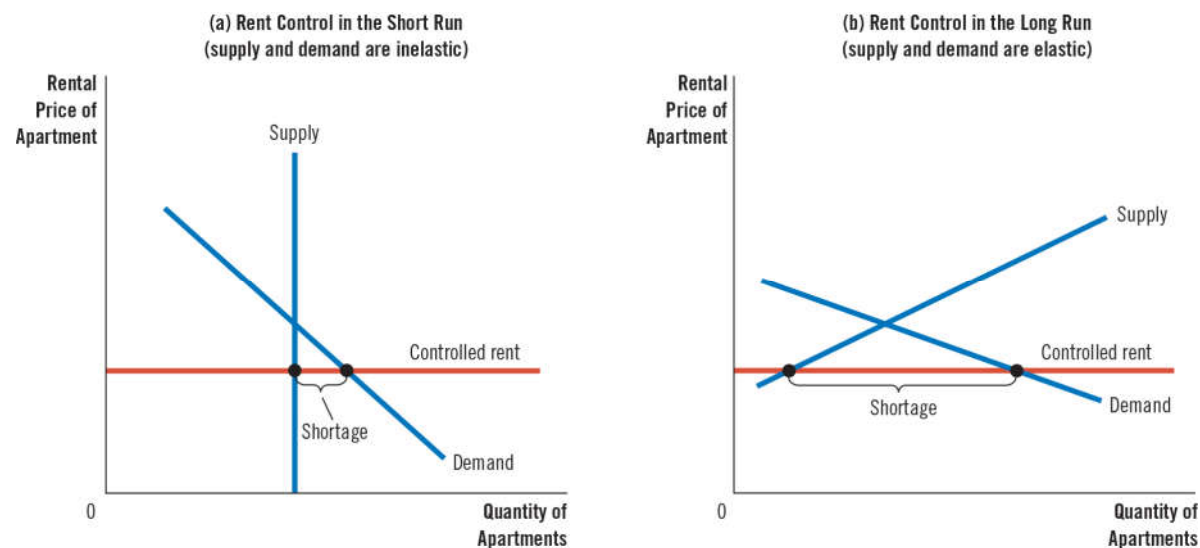
“Local ordinances that limit rent increases for some rental housing units, such as in New York and San Francisco, have had a positive impact over the past three decades on the amount and quality of broadly affordable rental housing in cities that have used them.”



Source: IGM Economic Experts Panel, February 7, 2012.

Figure 3 Rent Control in the Short Run and in the Long Run

Panel (a) shows the short-run effects of rent control: Because the supply and demand curves for apartments are relatively inelastic, the price ceiling imposed by a rent-control law causes only a small shortage of housing. Panel (b) shows the long-run effects of rent control: Because the supply and demand curves for apartments are more elastic, rent control causes a larger shortage.



How Price Floors Affect Market Outcomes

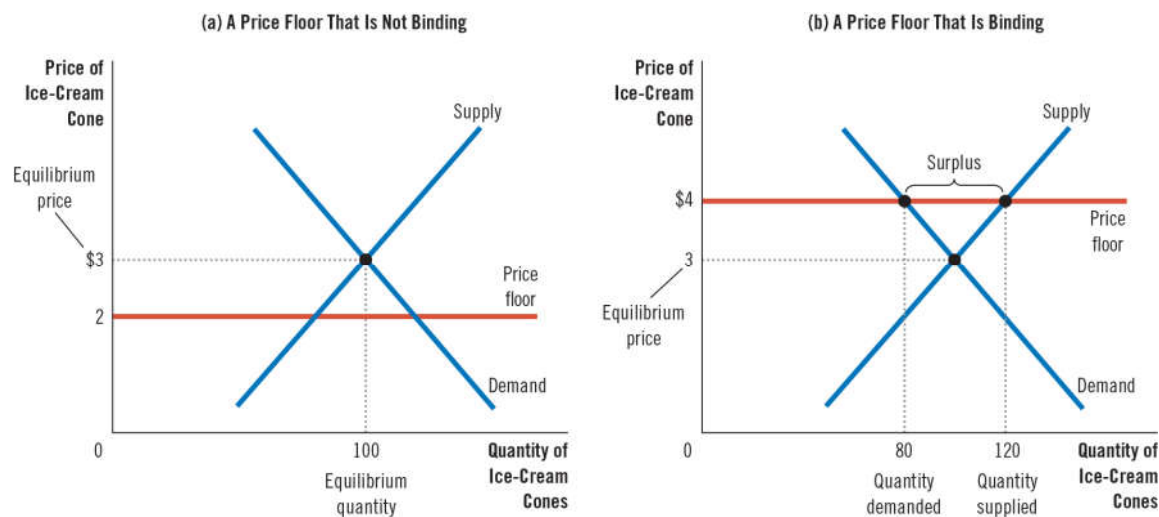
- Not binding
 - Set below the equilibrium price
 - No effect on price or quantity sold
- Binding constraint
 - Set above the equilibrium price
 - Some sellers are unable to sell what they want

Binding Price Floor

- When the government imposes a binding price ceiling
 - Surpluses arise
 - Sellers who appeal to the buyers' personal biases may be better able to sell their goods than those who do not
 - In a free market, price is the rationing mechanism
 - Sellers may not be happy about how much they are paid at the equilibrium price, but they can sell all they want

Figure 4 A Market with a Price Floor

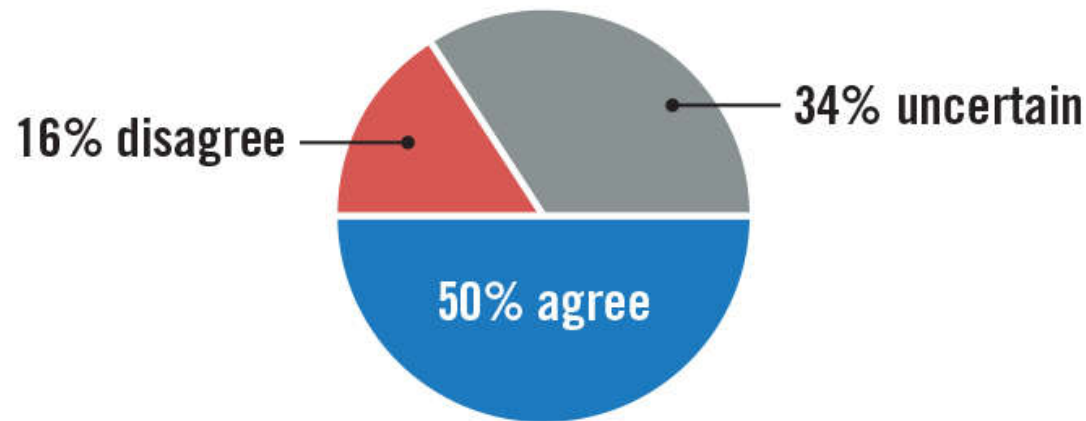
In panel (a), the government imposes a price floor of \$2. Because it is below the equilibrium price of \$3, the floor has no effect, and the market can reach the equilibrium of supply and demand. At this point, quantity supplied and demanded both equal 100 cones. In panel (b), the government imposes a price floor of \$4. Because the floor is above the equilibrium price of \$3, the market price is \$4. At this price, 120 cones are supplied while only 80 are demanded, so there is a surplus of 40 cones.



Ask the Experts: The Minimum Wage

“The current US federal minimum wage is \$7.25 per hour. States can choose whether to have a higher minimum—and many do. A federal minimum wage of \$15 per hour would lower employment for low-wage workers in many states.”

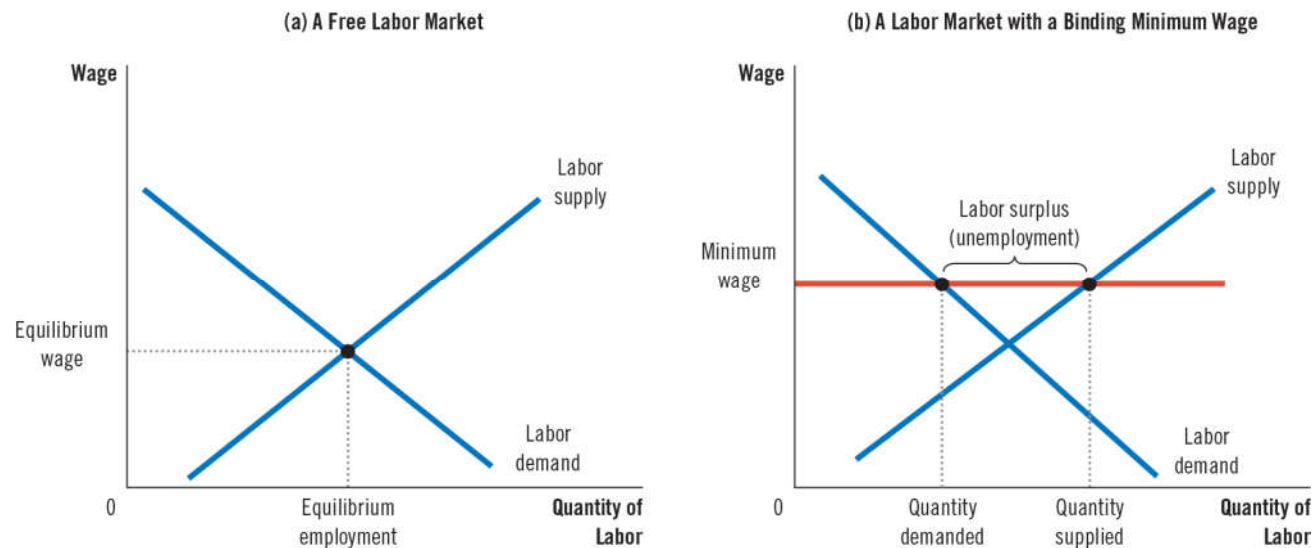
What do economists say?



Source: IGM Economic Experts Panel, February 2, 2021.

Figure 5 How the Minimum Wage Affects a Competitive Labor Market

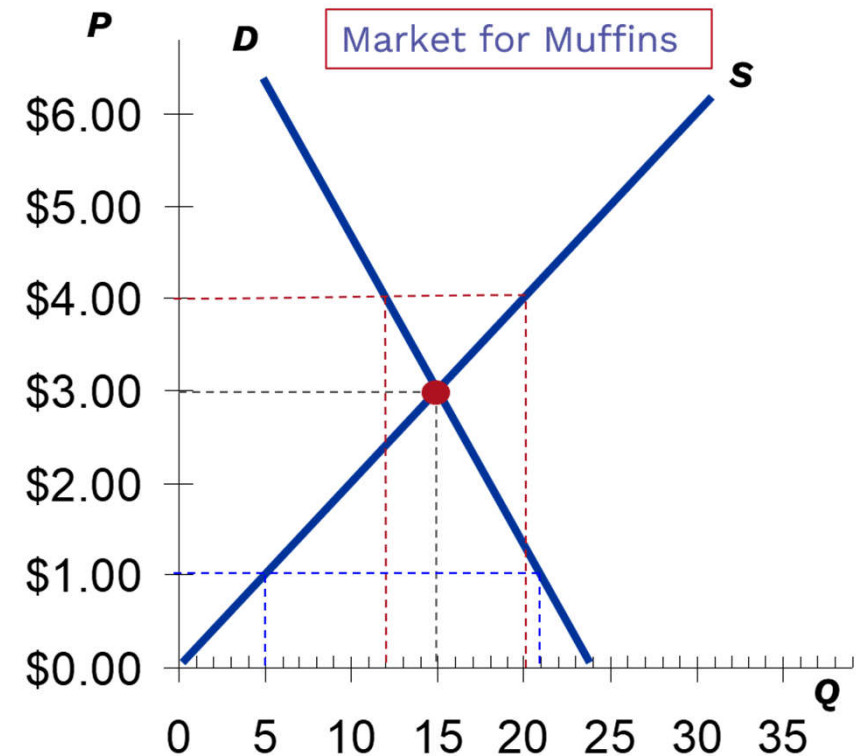
Panel (a) shows a labor market in which the wage adjusts to balance labor supply and labor demand. Panel (b) shows the impact of a binding minimum wage. Because the minimum wage is a price floor, it causes a surplus: The quantity of labor supplied exceeds the quantity demanded. The result is unemployment.



Active Learning 2: Price Floors for Muffins

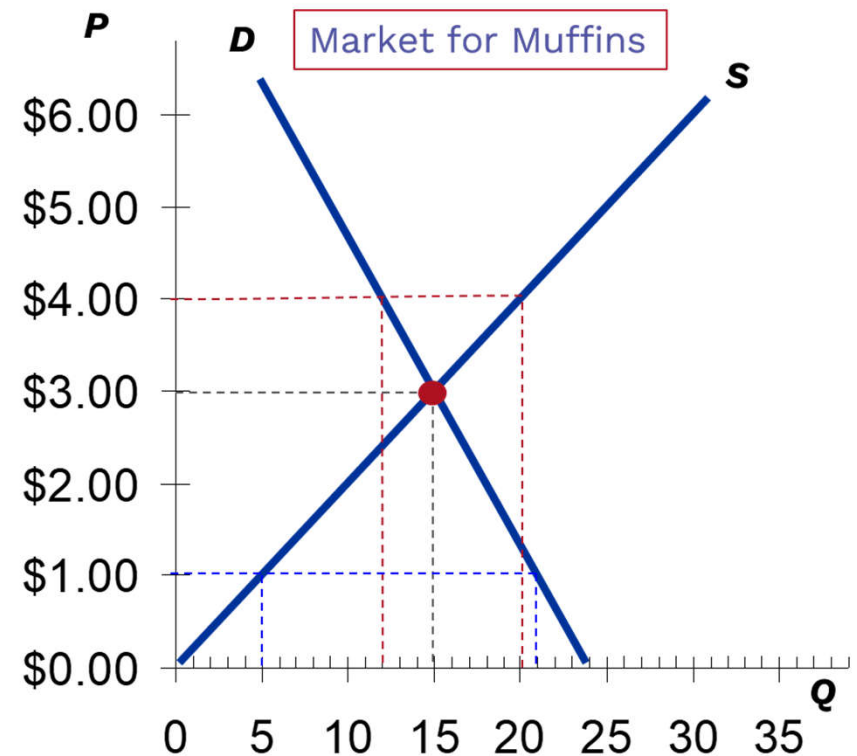
The Muffin Buyers' Association lobbies the government to impose a price floor. Which of the following is binding and what's the effect on the market?

- A. The price floor is set at \$1
- B. The price ceiling is set at \$4



Active Learning 2: Answers

- The price floor is set at \$1
 - Not binding
 - $P = \$3, Q = 15$
- The price floor is set at \$4
 - Binding
 - $P = \$4, Q_d = 12, Q_s = 20$
 - Surplus = 8 muffins



Evaluating Price Controls (1 of 2)

- Markets are usually a good way to organize economic activity
 - Prices balance supply and demand
 - Price-setting obscures signals that guide the allocation of society's resources
- Governments can sometimes improve market outcomes
 - Motivated to control prices
 - View the market's outcome as unfair

Evaluating Price Controls (2 of 2)

- Price controls can hurt some people they are intended to help
- Alternative policies are often better
 - Rent subsidy
 - Wage subsidy (earned income tax credit)

6.2 The Surprising Study of Tax Incidence

Taxes

- Governments use taxes to raise revenue for public projects
- Tax incidence*
 - Manner in which the burden of a tax is shared among participants in a market

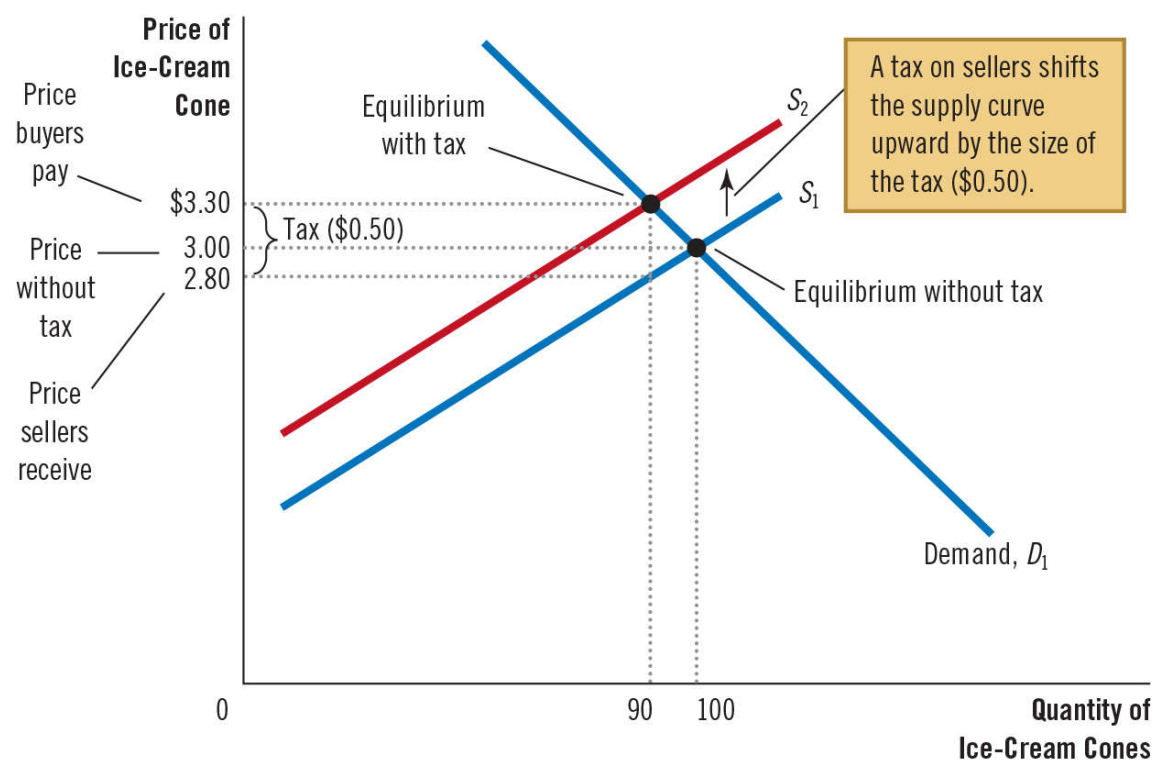
*Words accompanied by an asterisk are key terms from the chapter.

How Taxes on Sellers Affect Market Outcomes

- Taxes discourage market activity
 - Quantity sold is smaller in the new equilibrium
- Buyers and sellers share the tax burden
 - Buyers pay more, and sellers receive less
 - Sellers send the money to the government

Figure 6 A Tax on Sellers

- When a tax of \$0.50 is levied on sellers, the supply curve shifts up by \$0.50 from S_1 to S_2 .
- The equilibrium quantity falls from 100 to 90 cones.
- The price that buyers pay rises from \$3.00 to \$3.30.
- The price that sellers receive (after paying the tax) falls from \$3.00 to \$2.80.
- Even though sellers are legally responsible for paying the tax, buyers and sellers share the burden.

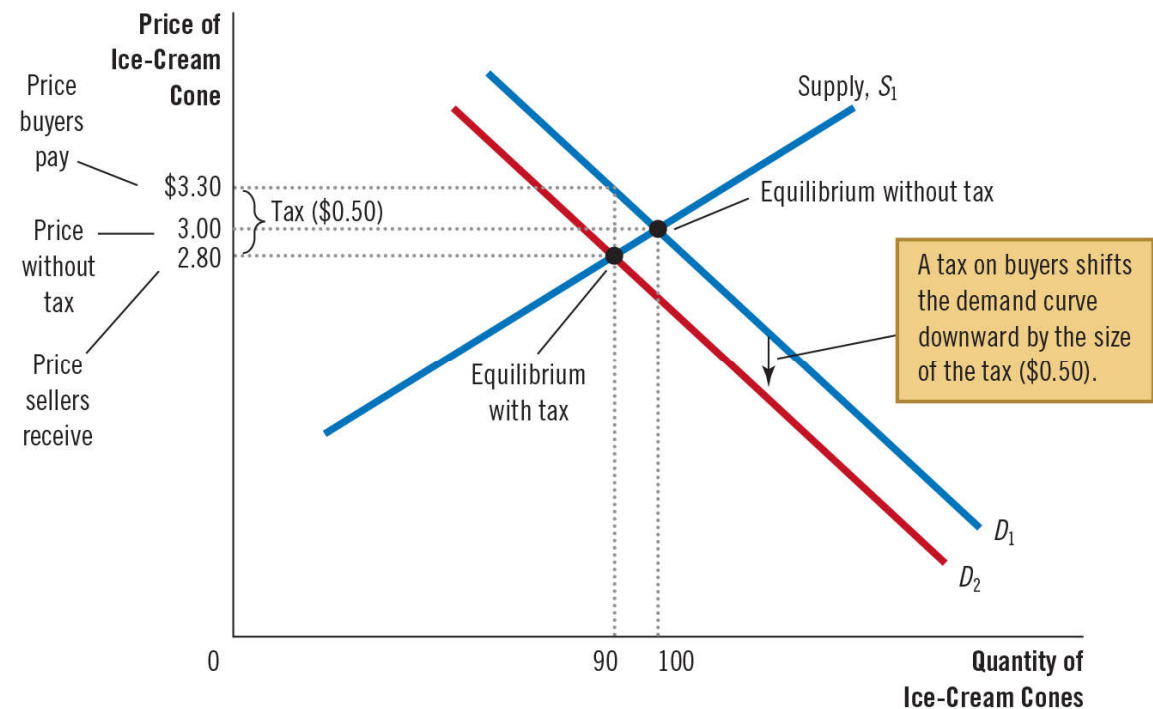


How Taxes on Buyers Affect Market Outcomes

- Taxes discourage market activity
 - Quantity sold is smaller in the new equilibrium
- Buyers and sellers share the tax burden
 - Buyers pay more, and sellers receive less
 - Buyers pay a lower market price but effective price (with tax) rises

Figure 7 A Tax on Buyers

- When a tax of \$0.50 is imposed on buyers, the demand curve shifts down by \$0.50 from D_1 to D_2 .
- The equilibrium quantity falls from 100 to 90 cones.
- The price that sellers receive falls from \$3.00 to \$2.80.
- The price that buyers pay (including the tax) rises from \$3.00 to \$3.30.
- Even though buyers are legally responsible for paying the tax, buyers and sellers share the burden.

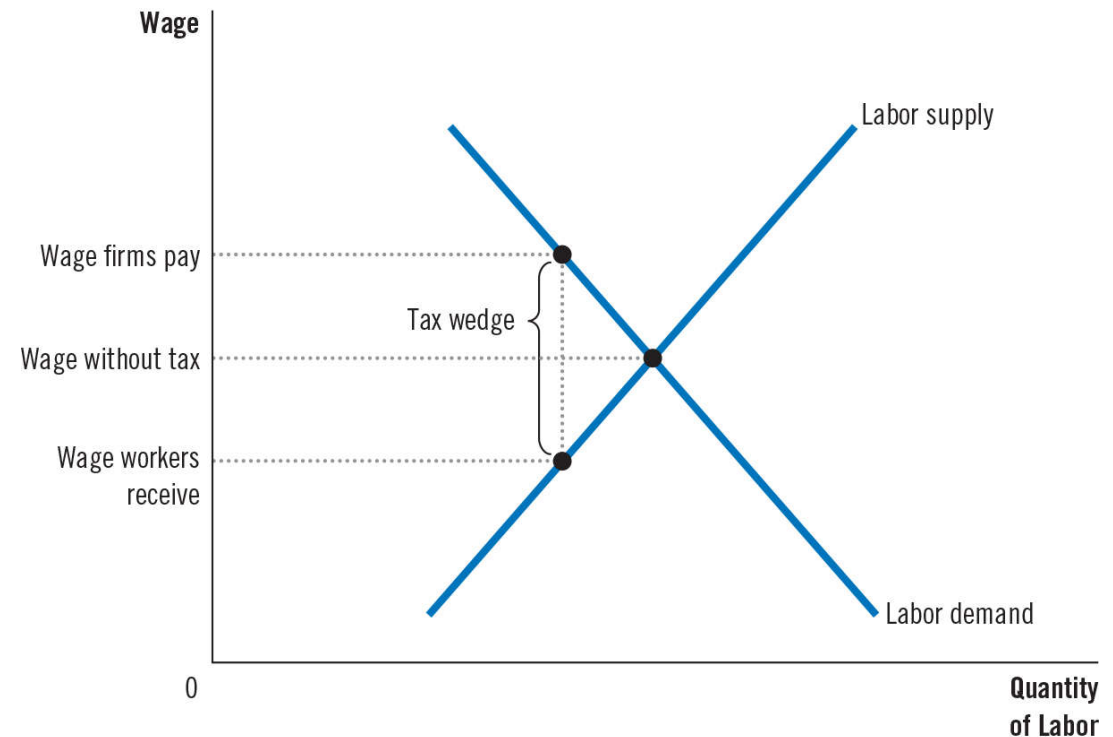


Tax Incidence

- Taxes on sellers and taxes on buyers are equivalent
- The tax
 - Inserts a wedge between price that buyers pay and price that sellers receive
 - Shifts the relative position of the supply and demand curves
 - Buyers and sellers share the tax burden in the new equilibrium

Figure 8 A Payroll Tax

- A payroll tax places a wedge between what firms pay and what workers receive.
- Comparing wages with and without the tax makes it clear that workers and firms share the tax burden.
- This division does not depend on whether the government imposes the tax entirely on workers, imposes it entirely on firms, or divides it equally between the two groups.



Elasticity and Tax Incidence (1 of 2)

- Tax burden falls more heavily on the side of the market that is less elastic
 - Elasticity measures the willingness of buyers or sellers to leave the market when conditions worsen
 - A small elasticity of demand means that buyers do not have good alternatives to consuming this particular good
 - A small elasticity of supply means that sellers do not have good alternatives to producing this particular good
 - The side of the market less willing to leave the market bears more of the burden of the tax

Elasticity and Tax Incidence (2 of 2)

- Elastic supply, inelastic demand
 - Price received by sellers does not fall much
 - Price paid by buyers rises substantially
 - Buyers bear most of the tax burden
- Inelastic supply, elastic demand
 - Price paid by buyers does not rise much
 - Price received by sellers falls substantially
 - Sellers bear most of the tax burden

Figure 9 How a Tax Burden Is Divided (1 of 2)

- In panel (a), the supply curve is elastic, and the demand curve is inelastic.
- In this case, the price received by sellers falls only slightly, while the price paid by buyers rises substantially. This means that buyers bear most of the tax burden.

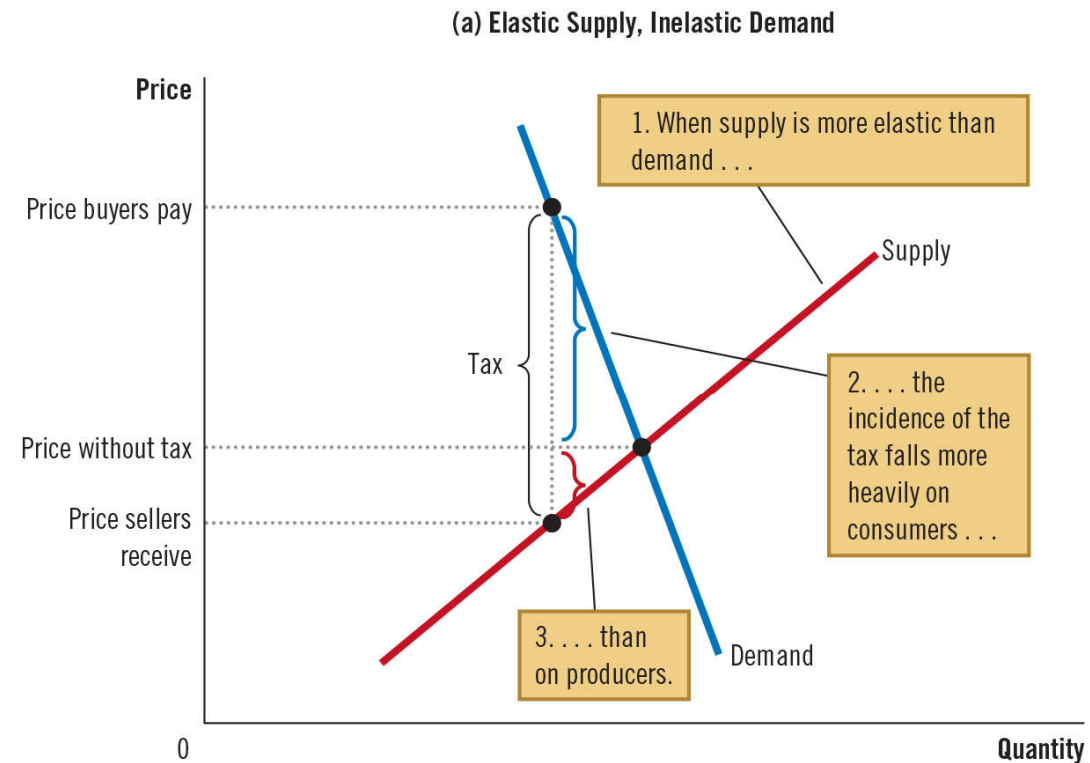
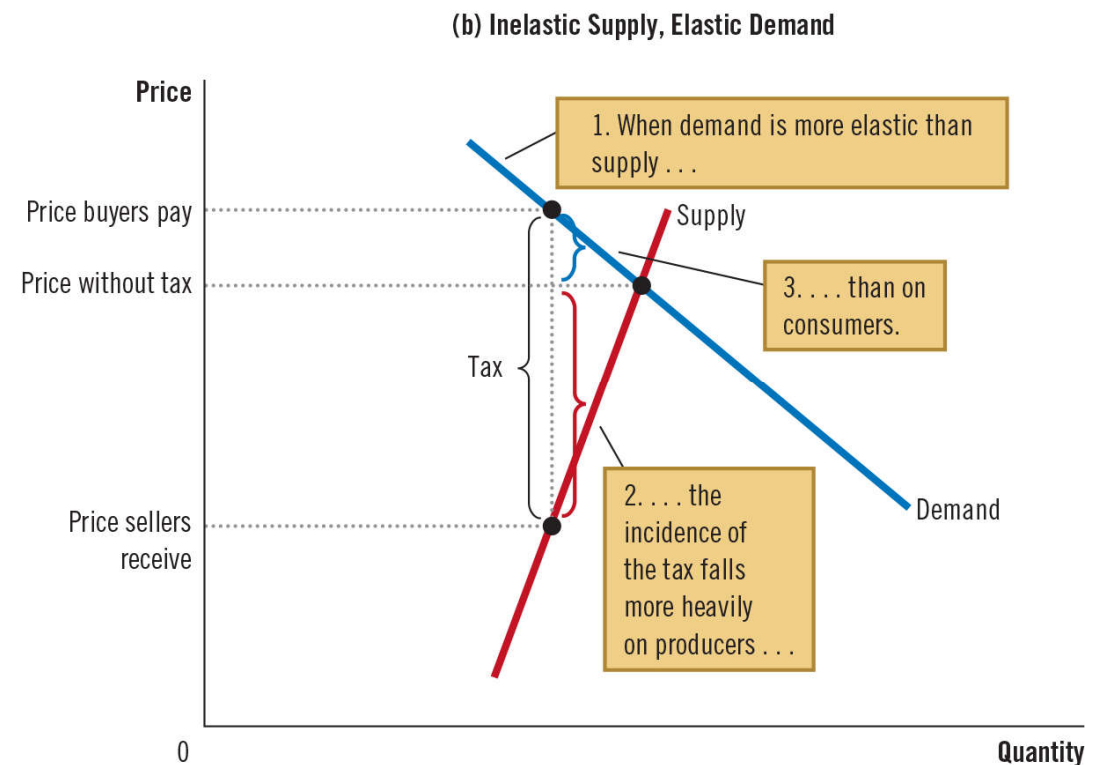


Figure 9 How a Tax Burden Is Divided (2 of 2)

- In panel (b), the situation is reversed: The supply curve is inelastic, and the demand curve is elastic.
- In this case, the price received by sellers falls substantially, while the price paid by buyers rises only slightly. Here, sellers bear most of the burden.



6.3 Conclusion

- Price controls and taxes are common in various markets, and their effects are frequently debated
- When analyzing government policies, supply and demand are the first and most useful tools of analysis

Think-Pair-Share Activity

Suppose that your state needs to raise more tax revenue. The governor proposes a tax on food because everyone must eat and, thus, a food tax would surely raise a great deal of tax revenue. The governor insists the tax should be placed on food sellers to protect less wealthy individuals who spend a large proportion of their income on food.

- A. Will the burden of a food tax fall only on the sellers of food as the governor said? Explain.
- B. Who will bear most of this tax burden? Explain.